

Illustrative scheduler decision — catalog manifests

$$S(m) = F_{cap} - \alpha M_{cost} - \beta L_{load} - \gamma E_{cost}$$

$$- \delta P_{evict} + \eta F_{fut} \quad [\alpha = 0.20, \beta = 0.25, \gamma = 0.10, \delta = 0.30, \eta = 0.35]$$

Stage: math derivation | $B_{RAM} = 8.0$ GB | Resident: research-expert

Candidate	Cap	RAM	Load	Enrg	Evict	Fut	Total	
math-expert	+0.91	-0.12	-0.07	-0.03	-0.02	+0.28	+0.95	SELECTED
coding-expert	+0.48	-0.15	-0.08	-0.03	-0.04	+0.10	+0.28	AVAIL
research-expert	+0.62	-0.16	-0.09	-0.04	-0.02	+0.18	+0.49	AVAIL
gen-reasoner	+0.85	-0.99	-0.30	-0.12	-0.15	+0.35	-0.36	OOM

general-reasoner rejected: RAM requirement (7.1 GB) exceeds available headroom (4.9 GB).